

UCS503P – Software Engineering Project

SkinSense

A Photo-Based Skin Analysis and Personalised Skincare Recommendation System

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Introduction

1.1 Background and Context

Skin conditions, and acne in particular, are among the most common dermatological concerns encountered across adolescent and adult populations. Diagnosis and product guidance are conventionally the domain of a dermatologist, yet routine access to professional skin assessment is often limited by cost, appointment availability, and geographic access to a specialist. At the same time, the consumer skincare market offers an overwhelming number of products, each formulated from combinations of active ingredients whose suitability depends on an individual’s skin type, existing sensitivities, and the severity of any condition present.

Without objective guidance, users frequently choose products through trial and error, informal advice, or marketing claims rather than an assessment of their own skin. SkinSense addresses this gap through a full-stack web application that accepts a user’s facial photograph, validates that the photograph is suitable for analysis, classifies skin type and acne severity using trained deep-learning models, and combines the analysis with declared allergies and budget constraints to produce a personalised, rule-filtered skincare routine.

Advances in computer vision and deep learning have made camera-based skin assessment practical. Residual networks such as ResNet [HZRS16] and efficient architectures such as MobileNetV2 [SHZ+18] provide suitable foundations for image classification, while MediaPipe [LTN+19] provides face-detection and landmark capabilities useful for image-quality validation.

1.1.1 Problem Statement

The project addresses three related gaps in the skincare decision-making process:

1. **Lack of objective self-assessment:** users need an accessible way to obtain a consistent image-based estimate of skin type and acne severity.
2. **Information overload in product selection:** users need assistance cross-checking products against skin type, allergies, ingredient conflicts, and budget.
3. **No clear signal for referral:** severe or low-confidence cases should be flagged for professional dermatological consultation rather than being treated as ordinary product-selection cases.

SkinSense is not intended to provide a medical diagnosis. It is a decision-support and recommendation system, and its analysis and referral limitations are communicated to the user.

1.2 Project Objectives

1. Implement MediaPipe-based photo validation for face presence, framing, lighting, and angle quality.
2. Train and integrate a ResNet-50 acne-severity classifier with three severity classes and a project test-accuracy target of at least 80%.
3. Develop a MobileNetV2-based skin-type classifier.
4. Implement a recommendation engine that filters products by skin profile and severity, removes declared allergens and conflicting ingredients, applies a budget constraint, and builds an AM/PM routine.
5. Implement JWT-based authentication and access control for user-specific analysis and preference data.
6. Deliver an end-to-end React/Vite frontend, FastAPI backend, machine-learning inference layer, and relational database integration.

Methodology and Design

2.1 System Architecture

SkinSense follows a three-tier web architecture consisting of a browser-based frontend, a REST API backend, and a relational data store, with a machine-learning inference layer invoked by the backend. This separation allows the trained models to be retrained or replaced without changing the frontend contract.

The principal workflow is represented through the use case, activity, class, collaboration, sequence, and data-flow diagrams included in this chapter. The four DFD artefacts are retained separately because they represent the context diagram, system-level decomposition, skin-analysis decomposition, and recommendation decomposition respectively.

2.2 Use Case Diagram

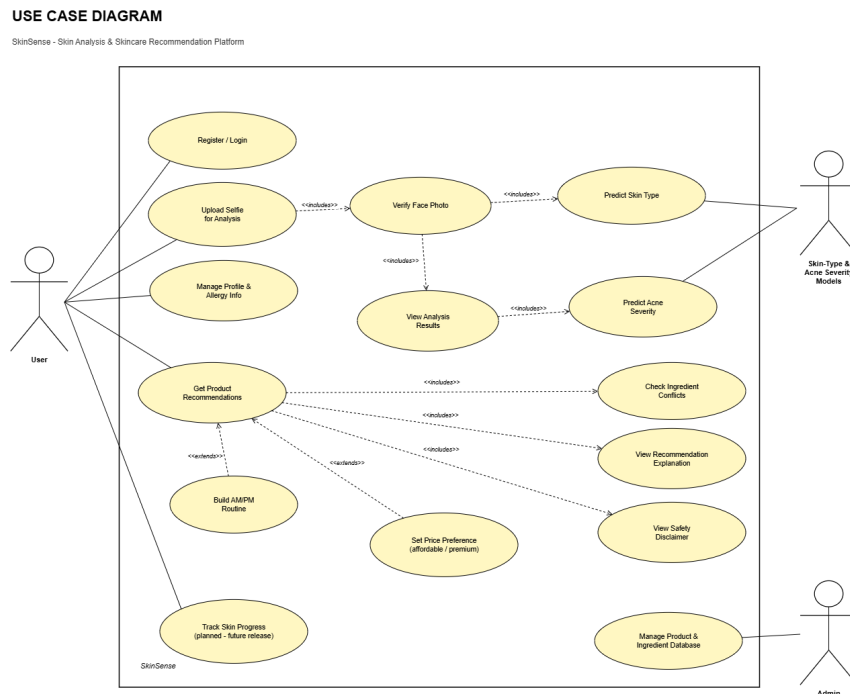


Figure 2.1: Use case diagram for the SkinSense system.

2.3 Use Case Tables

SkinSense : Use Case Specifications

Skin Analysis & Skincare Recommendation Platform

15 use cases, one per ellipse in the Use Case Diagram. Legend: AA = Actor Action, SR = System Response, SA = System Action. “Type” and all <<includes>>/<<extends>> relationships below are taken directly from the diagram's actor-association lines and dependency arrows.

UC-01	Register / Login		
Precondition	None		
Use Case	Register / Login		
Actors	User		
Purpose	Let a user create an account or sign in, so their profile and scan history are attached to them.		
Overview	The user provides credentials; the system authenticates or creates an account and starts a session.		
Type	Primary (direct association with User)		
Typical Course of Events	Step	By	Description
	1	SR	The system displays a login/register form.
	2	AA	The user enters credentials (or chooses to register).
	3	SR	The system validates the credentials against stored records.
	4	SR	The system starts an authenticated session and redirects to the Upload page.
Alternate Flow	Step	Description	
	2	The user chooses “Continue without an account.” The system proceeds with an anonymous session; profile settings are stored locally only (see UC-08).	
Exceptional Flow	Step	Description	
	3	Credentials do not match any record. The system displays an error and re-prompts.	
Post-condition	An authenticated (or anonymous) session exists for subsequent use cases.		

UC-02	Upload Selfie for Analysis
Precondition	<i>User has a session (UC-01).</i>
Use Case	<i>Upload Selfie for Analysis</i>
Actors	<i>User</i>
Purpose	<i>Let the user submit a face photo as the input to skin analysis.</i>

Overview	The user selects or drags in an image file, which the system stores and forwards for validation.		
Type	Primary (direct association with User)		
Typical Course of Events	Step	By	Description
	1	SR	The system presents an upload control (file picker or drag-and-drop).
	2	AA	The user selects a JPG or PNG photo of their face.
	3	SR	The system shows a preview of the selected image.
	4	AA	The user confirms and submits the photo.
	5	SR	The system stores the image and assigns it an image_id.
	6	SR	The system triggers UC-03: Verify Face Photo (<<includes>>).
Alternate Flow	Step	Description	
	2	The user instead chooses “Choose manually” and proceeds directly to entering skin type and severity by hand, bypassing this use case entirely.	
Exceptional Flow	Step	Description	
	2	The selected file is not an image, or exceeds 10 MB. The system rejects the selection with a specific reason and re-prompts.	
Post-condition	A stored image with an image_id exists, ready for validation.		

UC-03	Verify Face Photo		
Precondition	<i>A photo has been uploaded (UC-02).</i>		
Use Case	<i>Verify Face Photo</i>		
Actors	<i>User (indirectly, as the subject of validation)</i>		
Purpose	<i>Reject unusable photos before they reach the ML models, with a specific, actionable reason.</i>		
Overview	<i>MediaPipe checks face count, head angle, brightness, and blur on the uploaded image.</i>		
Type	<i>Secondary (<<includes>> from UC-02)</i>		
Typical Course of Events	Step	By	Description
	1	SR	<i>The system runs face detection on the stored image.</i>
	2	SR	<i>The system checks exactly one face is present, head yaw is within range, brightness is adequate, and blur is below threshold.</i>
	3	SR	<i>If all checks pass, the system proceeds to UC-04: Predict Skin Type and UC-05: View Analysis Results (<<includes>>).</i>

Exceptional Flow	Step	Description
	2	One or more checks fail. The system returns the specific failing reasons (e.g. "No face detected", "Photo is too dark", "Face is turned too far to the side"), then prompts the user to retake the photo or switch to manual entry.
Post-condition	The photo is confirmed usable, or the user has specific, actionable rejection reasons.	

UC-04	Predict Skin Type		
Precondition	Photo has passed validation (UC-03).		
Use Case	Predict Skin Type		
Actors	Skin-Type & Acne Severity Models		
Purpose	Classify the user's skin as oily, dry, or normal from their photo.		
Overview	A MobileNetV2 classifier (ImageNet transfer learning, frozen base, fine-tuned head) returns a 3-class prediction with a confidence score.		
Type	Secondary (<<includes>> from UC-03); direct association with Skin-Type & Acne Severity Models		
Typical Course of Events	Step	By	Description
	1	SR	The system passes the validated image to the skin-type model.
	2	SR	The model returns a label (oily / dry / normal) and a confidence score.
	3	SR	The system includes this result in UC-05: View Analysis Results.
Exceptional Flow	Step	Description	
	2	Confidence is below the low-confidence threshold (0.60). The system flags this for the referral decision in UC-05.	
Post-condition	skin_type and skin_type_confidence are set on the prediction result.		

UC-05	View Analysis Results		
Precondition	Photo has passed validation (UC-03), or skin type and severity were entered manually (alternate flow of UC-02).		
Use Case	View Analysis Results		
Actors	User		
Purpose	Show the user their skin type, acne severity, and any relevant referral advice before they proceed to recommendations.		
Overview	The system displays the two classifications, confidence levels, estimated lesion count, detected skin tone, and a dermatologist referral banner if warranted.		

Type	Secondary (<<includes>> from UC-03)		
Typical Course of Events	Step	By	Description
	1	SR	The system includes UC-06: Predict Acne Severity (<<includes>>).
	2	SR	The system computes the referral flag: needed if severity is severe, either confidence is below 0.60, or validation returned warnings.
	3	SR	The system displays skin type, acne severity, confidence, lesion count estimate, and detected skin tone/undertone.
	4	SR	If referral is needed, the system displays a banner recommending a dermatologist consultation with the specific reason(s).
	5	AA	The user reviews the results and proceeds to UC-07: Get Product Recommendations.
Alternate Flow	Step	Description	
	5	The user instead returns to UC-02 to try a different photo.	
Post-condition	The user has seen their classification and any referral advice.		

UC-06	Predict Acne Severity		
Precondition	<i>Invoked from UC-05, on a photo that has passed validation (UC-03).</i>		
Use Case	<i>Predict Acne Severity</i>		
Actors	<i>Skin-Type & Acne Severity Models</i>		
Purpose	<i>Grade the user's acne into mild, moderate, or severe from their photo.</i>		
Overview	<i>A ResNet-50 classifier, following Zhang & Ma's classification-module approach on the ACNE04 dataset, reclassifies severity by lesion count into three bands (mild 1–5, moderate 6–20, severe >20).</i>		
Type	<i>Secondary (<<includes>> from UC-05); direct association with Skin-Type & Acne Severity Models</i>		
Typical Course of Events	Step	By	Description
	1	SR	<i>The system passes the validated image to the acne-severity model.</i>
	2	SR	<i>The model returns a severity label, a confidence score, and an estimated lesion count.</i>
	3	SR	<i>The system returns this result to UC-05: View Analysis Results.</i>
Exceptional Flow	Step	Description	
	2	<i>Confidence is below threshold, or the lesion count is unusually high (dense acne, outside the model's validated range). The system flags this for the referral decision in UC-05.</i>	

Post-condition	<i>acne_severity, acne_severity_confidence, and lesion_count_estimate are set on the prediction result.</i>
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UC-07	Get Product Recommendations		
Precondition	Results have been viewed (UC-05).		
Use Case	Get Product Recommendations		
Actors	User		
Purpose	Produce a ranked, explainable set of skincare products matching the user's skin profile.		
Overview	A deterministic rule-based engine matches skin type and severity against the product/ingredient catalog, applying allergy and budget filters and excluding conflicting ingredient pairs.		
Type	Primary (direct association with User); <<extends>> from UC-09 and UC-13		
Typical Course of Events	Step	By	Description
	1	AA	The user requests recommendations, optionally supplying allergy exclusions and a budget ceiling.
	2	SR	The system matches products against skin type and acne severity.
	3	SR	The system includes UC-10: Check Ingredient Conflicts (<<includes>>).
	4	SR	The system includes UC-11: View Recommendation Explanation (<<includes>>).
	5	SR	The system includes UC-12: View Safety Disclaimer (<<includes>>).
	6	SR	The system returns the ranked product list with reasons, plus an excluded list with reasons for every filtered-out product.
	7	SR	The system triggers UC-09: Build AM/PM Routine.
Exceptional Flow	Step	Description	
	2	No products satisfy the filters. The system returns an empty product list with a note explaining why (e.g. "Try raising the budget or removing an ingredient").	
Post-condition	A recommendation result (products, exclusions, routine) is available to display.		

UC-08	Manage Profile & Allergy Info
Precondition	<i>User has a session (UC-01).</i>
Use Case	<i>Manage Profile & Allergy Info</i>

Actors	<i>User</i>		
Purpose	<i>Let the user store ingredients to avoid and a budget ceiling once, rather than re-entering them every scan.</i>		
Overview	<i>A profile page collects allergy/exclusion ingredients, a per-product budget, and free-text notes, persisted locally and applied automatically on future scans.</i>		
Type	<i>Primary (direct association with User)</i>		
Typical Course of Events	Step	By	Description
	1	AA	<i>The user opens the Skin Profile page.</i>
	2	AA	<i>The user enters ingredients to avoid, a budget ceiling, and any notes.</i>
	3	SR	<i>The system saves the profile.</i>
	4	SR	<i>On the next scan, the system pre-fills UC-07's filters from this saved profile.</i>
Post-condition	<i>The user's allergy and budget preferences are persisted for future use.</i>		

UC-09	Build AM/PM Routine		
Precondition	<i>Products have been matched (UC-07).</i>		
Use Case	<i>Build AM/PM Routine</i>		
Actors	<i>User</i>		
Purpose	<i>Organize recommended products into a usable morning and evening sequence.</i>		
Overview	<i>The system orders products into the canonical step sequence (cleanser → toner → treatment → serum → moisturizer → sunscreen), keeps sunscreen AM-only and retinoids PM-only, and keeps conflicting ingredients in separate halves of the day.</i>		
Type	<i>Secondary (<<extends>> UC-07)</i>		
Typical Course of Events	Step	By	Description
	1	SR	<i>The system assigns each recommended product to AM, PM, or both, based on its category and ingredients.</i>
	2	SR	<i>The system orders each half of the day by the canonical step sequence.</i>
	3	SR	<i>The system returns the ordered routine.am and routine.pm product-id lists.</i>
Post-condition	<i>An ordered AM and PM routine is attached to the recommendation result.</i>		

UC-10	Check Ingredient Conflicts		
Precondition	<i>Products have been matched (UC-07).</i>		

Use Case	<i>Check Ingredient Conflicts</i>		
Actors	<i>User (as beneficiary; no direct interaction)</i>		
Purpose	<i>Prevent two actively conflicting ingredients from being recommended together.</i>		
Overview	<i>Sourced conflict pairs (e.g. retinoid + AHA/BHA, retinoid + benzoyl peroxide, benzoyl peroxide + vitamin C) are checked against the matched product set.</i>		
Type	<i>Secondary (<<includes>> from UC-07)</i>		
Typical Course of Events	Step	By	Description
	1	SR	<i>The system checks the matched product set against the documented conflict-pair table.</i>
	2	SR	<i>Where a conflict is found, the system either separates the two products across AM/PM (if resolvable by timing) or excludes the lower-ranked one.</i>
	3	SR	<i>Every exclusion is recorded with its specific reason.</i>
Post-condition	<i>No two actively-conflicting ingredients appear in the same routine half.</i>		

UC-11	View Recommendation Explanation		
Precondition	<i>Products have been matched (UC-07).</i>		
Use Case	<i>View Recommendation Explanation</i>		
Actors	<i>User</i>		
Purpose	<i>Let the user understand why each product was recommended, not just that it was.</i>		
Overview	<i>Every recommended product carries at least one plain-language reason tied to the user's skin type, severity, or preferences.</i>		
Type	<i>Secondary (<<includes>> from UC-07)</i>		
Typical Course of Events	Step	By	Description
	1	SR	<i>The system attaches one or more reasons to each recommended product (e.g. "Non-comedogenic for oily skin", "Targets moderate acne").</i>
	2	AA	<i>The user reads the reasons alongside each product.</i>
Post-condition	<i>Every displayed product has a visible, specific reason for its inclusion.</i>		

UC-12	View Safety Disclaimer		
Precondition	<i>Products have been matched (UC-07).</i>		
Use Case	<i>View Safety Disclaimer</i>		
Actors	<i>User</i>		

Purpose	<i>Make clear that SkinSense is not a substitute for professional dermatological advice.</i>		
Overview	<i>A persistent disclaimer is shown alongside recommendations, and is emphasized further when a referral has been flagged (UC-05).</i>		
Type	<i>Secondary (<<includes>> from UC-07)</i>		
Typical Course of Events	Step	By	Description
	1	SR	<i>The system displays a standing disclaimer with the recommendations.</i>
	2	AA	<i>The user acknowledges by continuing to use the app.</i>
Post-condition	<i>The disclaimer has been shown alongside every set of recommendations.</i>		

UC-13	Set Price Preference (affordable / premium)		
Precondition	<i>None (may be set before or during UC-07).</i>		
Use Case	<i>Set Price Preference (affordable / premium)</i>		
Actors	<i>User</i>		
Purpose	<i>Let the user bias recommendations toward their preferred price tier without a hard budget cutoff.</i>		
Overview	<i>An optional preference that adjusts ranking toward affordable or premium products, distinct from the hard budget ceiling in UC-08.</i>		
Type	<i>Secondary (<<extends>> UC-07)</i>		
Typical Course of Events	Step	By	Description
	1	AA	<i>The user selects “affordable” or “premium” as a preference.</i>
	2	SR	<i>The system re-ranks the matched product set to favor the chosen tier.</i>
Post-condition	<i>The product ranking reflects the user's price preference.</i>		

UC-14	Track Skin Progress (planned — future release)		
Precondition	<i>Multiple scans exist for the user over time.</i>		
Use Case	<i>Track Skin Progress (planned — future release)</i>		
Actors	<i>User</i>		
Purpose	<i>Show the user how their skin type and acne severity have changed across repeated scans.</i>		
Overview	<i>Not part of the committed build; included in the diagram as a planned, future-release use case only, directly associated with the User actor for scope-completeness.</i>		

Type	<i>Primary (direct association with User) — planned, not scheduled in the current build</i>
Typical Course of Events	<i>Not applicable — this use case is not implemented in the current build.</i>
Post-condition	<i>Not applicable in the current build.</i>

UC-15	Manage Product & Ingredient Database		
Precondition	Admin has a session.		
Use Case	Manage Product & Ingredient Database		
Actors	Admin		
Purpose	Maintain the product catalog, ingredient tags, allergen list, and conflict-pair table that UC-07 and UC-10 depend on.		
Overview	An administrative interface (or direct data-file editing, in the current build) for adding, editing, or retiring catalog entries.		
Type	Primary (direct association with Admin)		
Typical Course of Events	Step	By	Description
	1	AA	The Admin adds, edits, or removes a product, ingredient, allergen, or conflict-pair entry.
	2	SR	The system validates the entry (e.g. every conflict pair must cite a reason) and persists it.
	3	SR	Subsequent runs of UC-07 and UC-10 use the updated catalog.
Exceptional Flow	Step	Description	
	2	The entry is malformed (e.g. a conflict pair with no reason, per the sourcing requirement). The system rejects the change and reports why.	
Post-condition	The product/ingredient database reflects the Admin's changes for all future recommendations.		

2.3 Activity Diagram

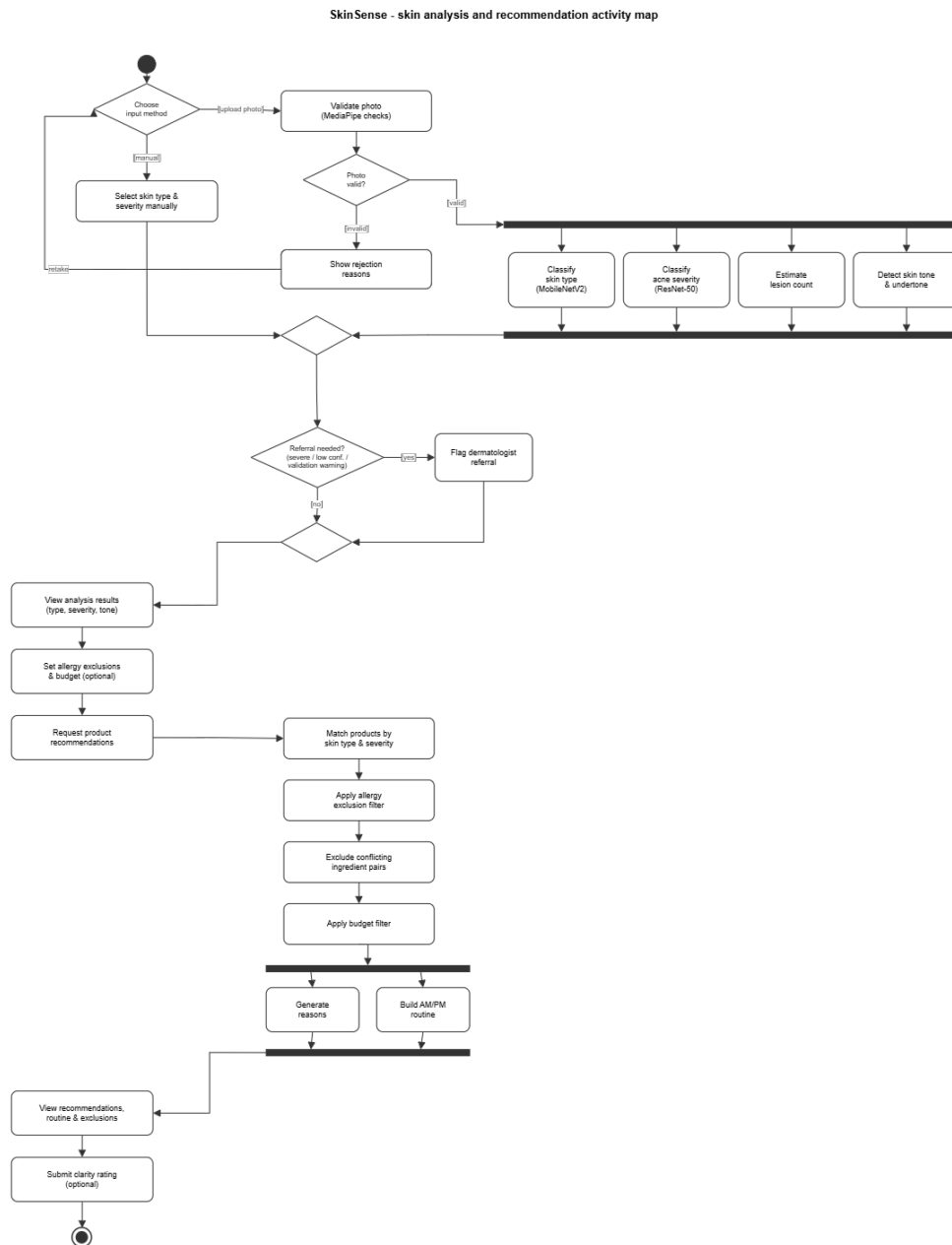


Figure 2.2: Activity diagram for the SkinSense user workflow.

2.4 Swimlane Activity Diagram

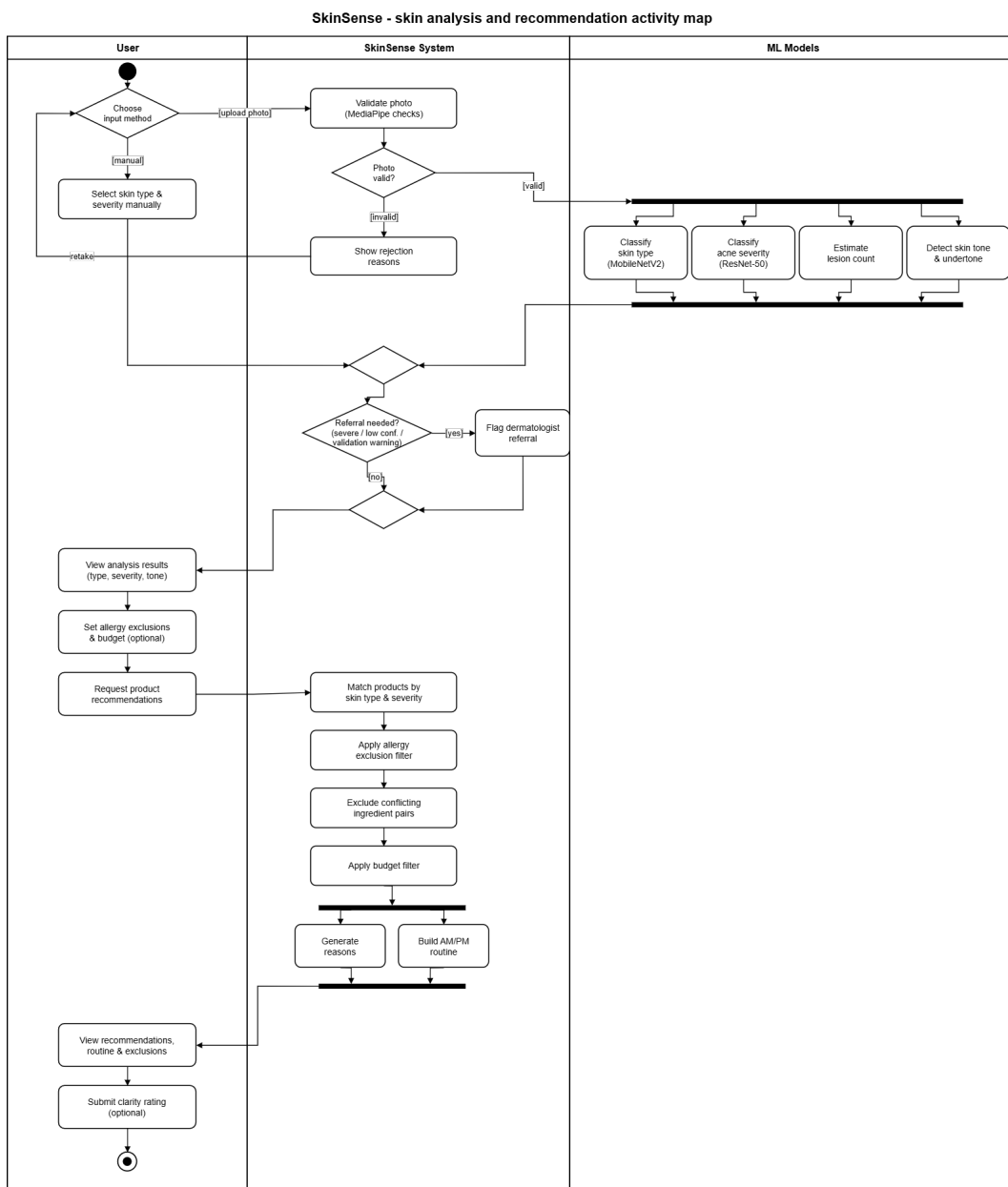
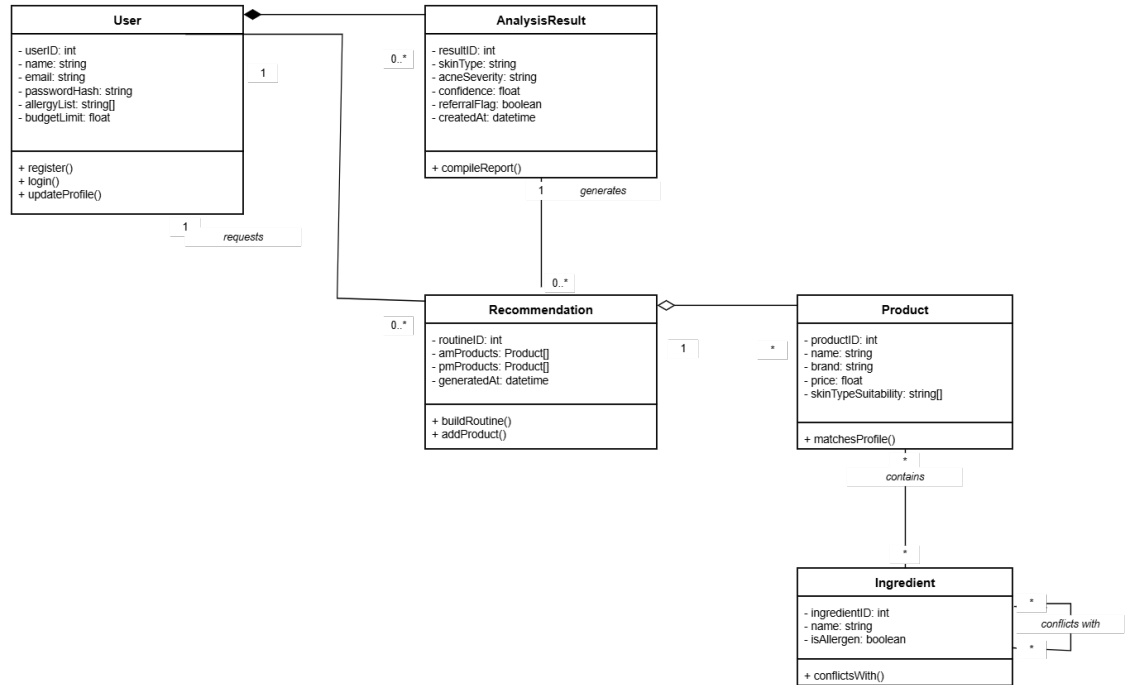


Figure 2.3: Swimlane activity diagram showing responsibilities of the User, SkinSense System, and ML Models.

2.5 Class Diagram

CLASS DIAGRAM — SkinSense Domain Model

Core persistent entities of the system and the relationships between them



Notation: filled diamond = composition, hollow diamond = aggregation, plain line = association (with role name and multiplicities). Attributes are private (-); operations are public (+).

Figure 2.4: Class diagram showing the core SkinSense entities and relationships.

2.6 Collaboration Diagram

COLLABORATION DIAGRAM — SkinSense Main User Journey

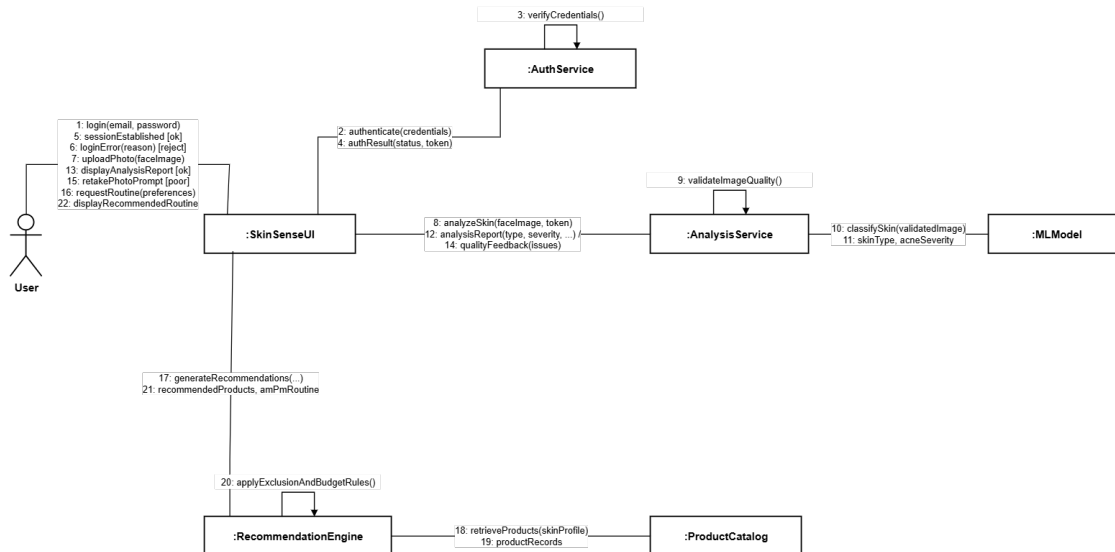


Figure 2.5: Collaboration diagram for the main SkinSense interaction sequence.

2.7 Sequence Diagram

SEQUENCE DIAGRAM : SkinSense Main User Journey

Scenario: a registered user logs in, submits a facial image for analysis, and receives a personalised skincare routine.

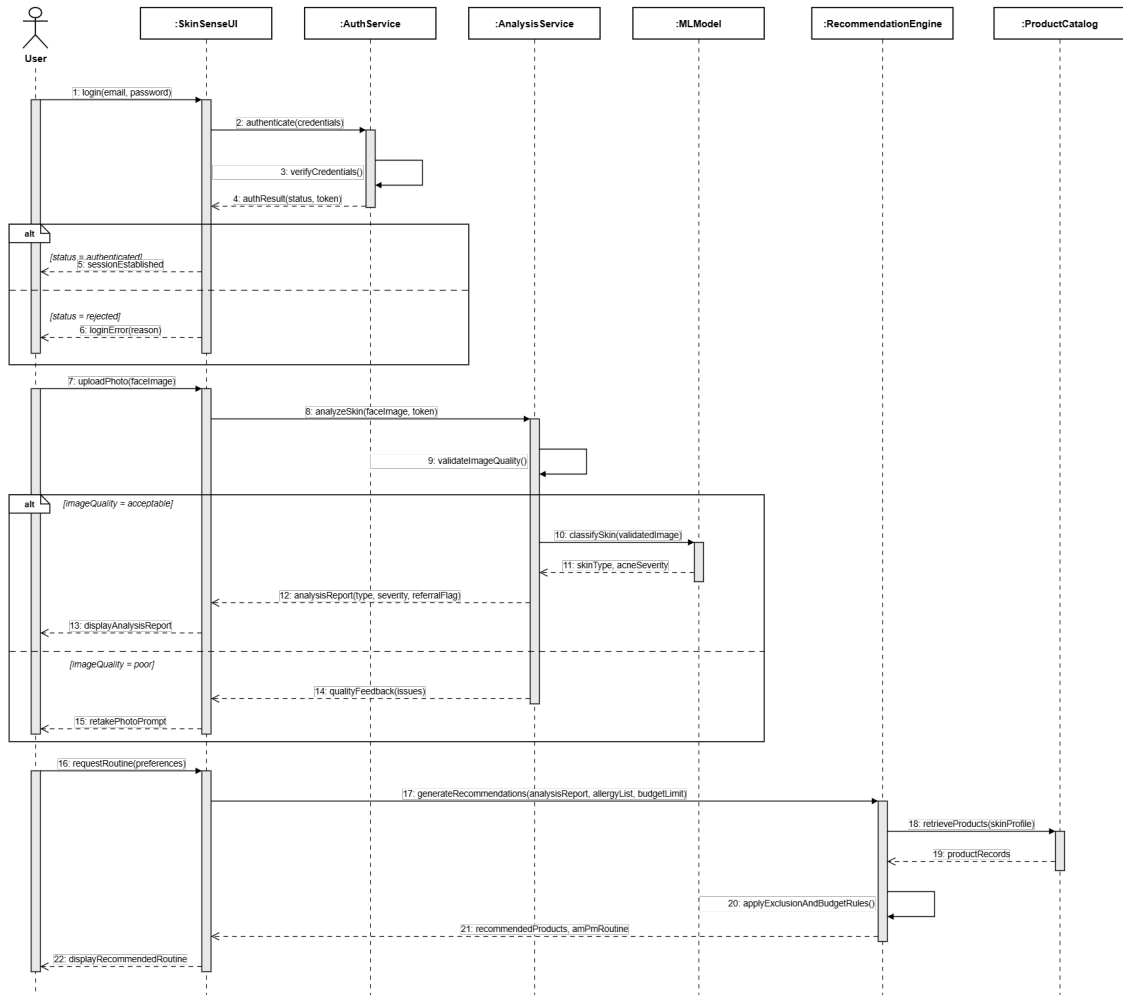


Figure 2.6: Sequence diagram for the main registered-user SkinSense journey.

2.8 Project Gantt Chart

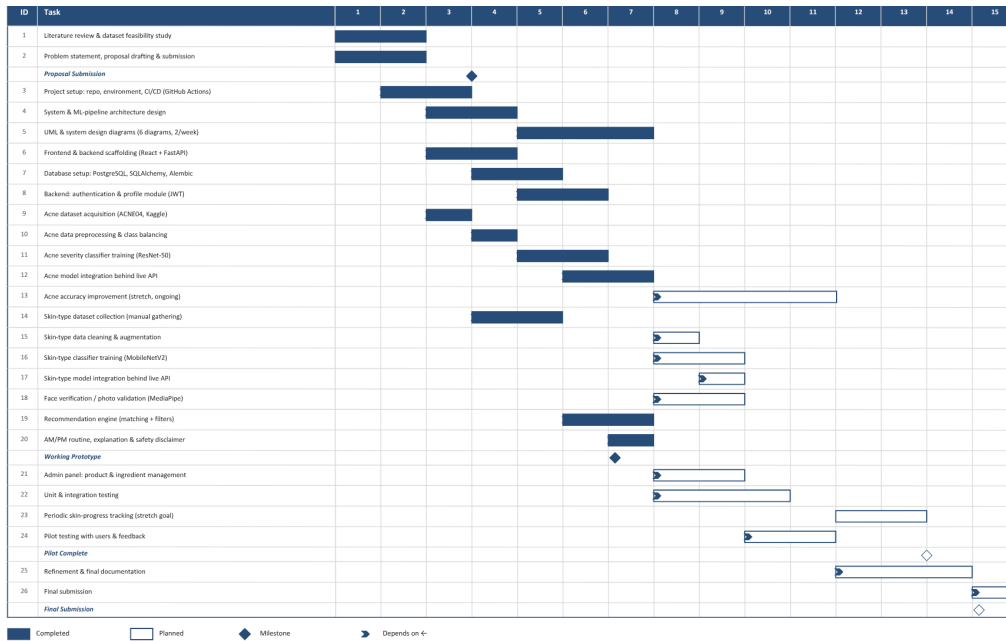


Figure 2.7: Gantt chart showing the planned project timeline and task schedule for SkinSense.

2.9 Data Flow Diagrams

2.9.1 DFD Level 0 – Context Diagram

LEVEL 0 DFD : CONTEXT DIAGRAM

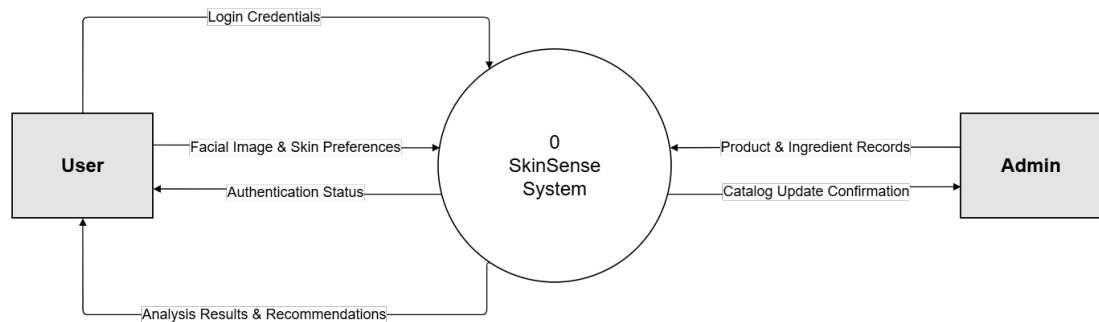


Figure 2.8: DFD Level 0 context diagram for SkinSense.

2.9.2 DFD Level 1 – System Decomposition

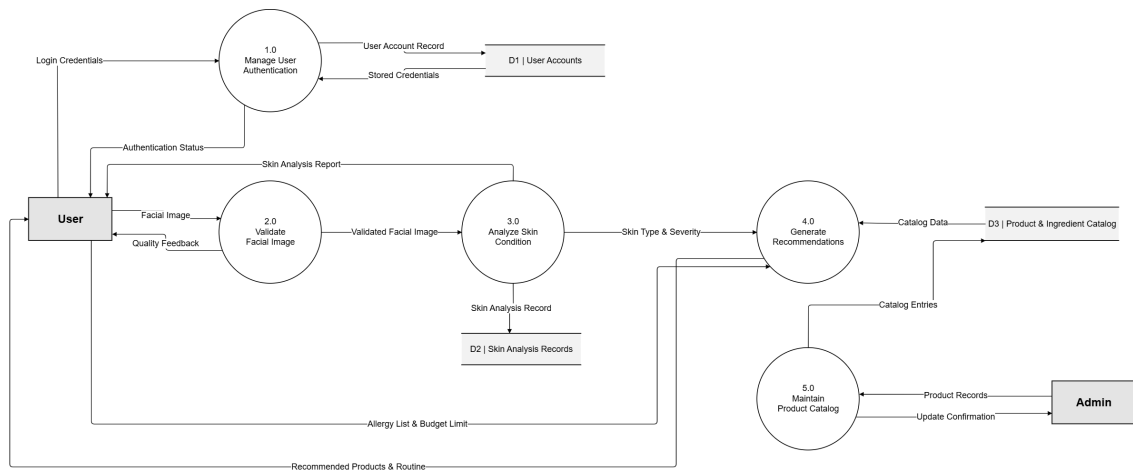


Figure 2.9: DFD Level 1 decomposition of the SkinSense system.

2.9.3 DFD Level 2 – Skin Analysis Decomposition

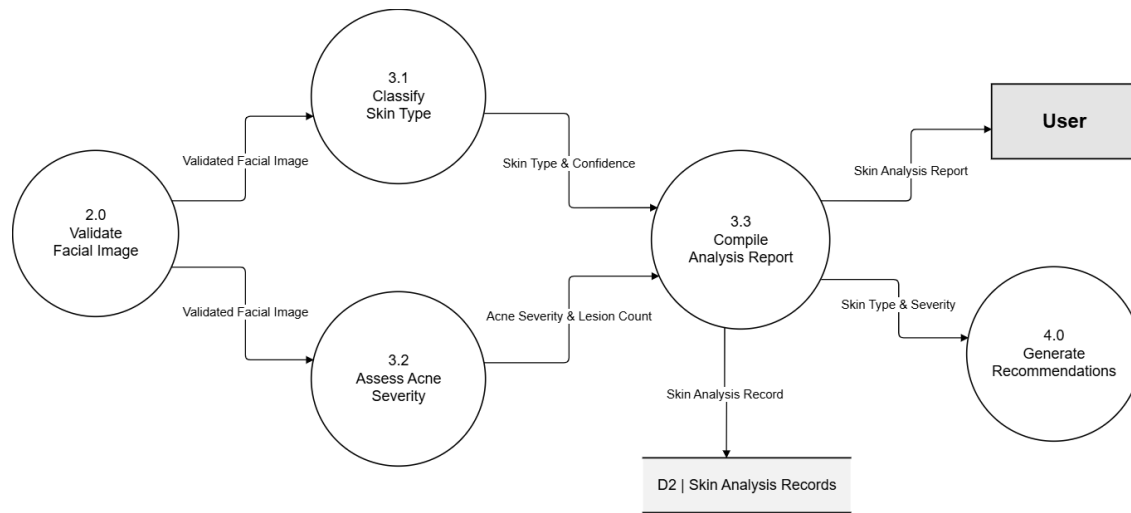


Figure 2.10: DFD Level 2 decomposition of the Analyze Skin Condition process.

2.9.4 DFD Level 2 – Recommendation Decomposition

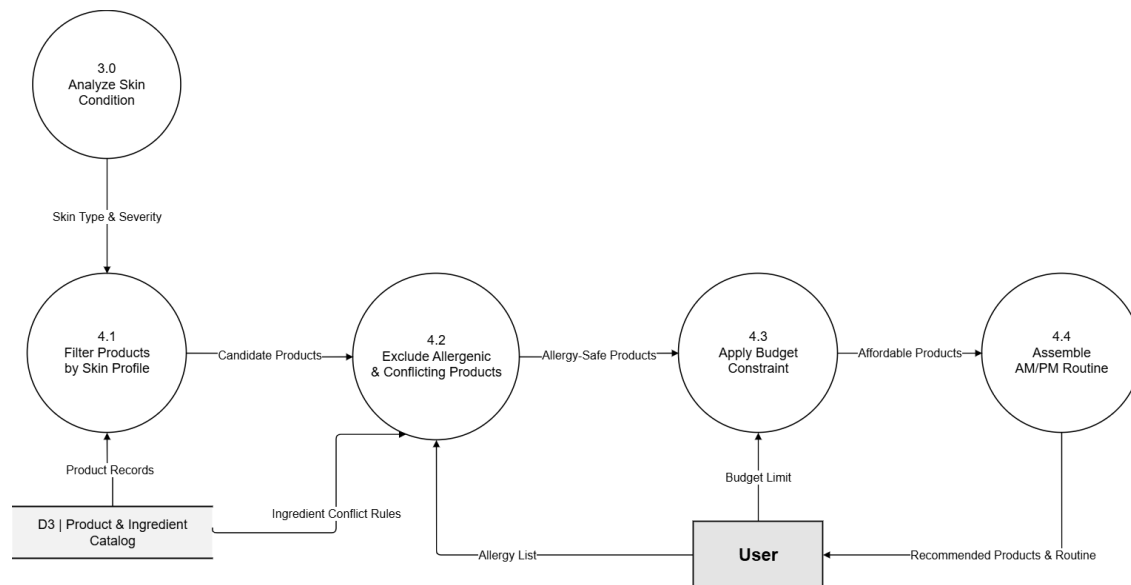


Figure 2.11: DFD Level 2 decomposition of the Generate Recommendations process.

2.9.5 DFD Naming Convention

The four supplied DFD images are retained as four distinct artefacts. Their filenames have been normalised into descriptive names corresponding to their level and process:

- `dfd_level_0_context.png`
- `dfd_level_1_overview.png`
- `dfd_level_2_analysis.png`
- `dfd_level_2_recommendation.png`

2.10 Technical Stack

- **Frontend:** React with Vite (JavaScript), using shadcn/ui components. The interface covers upload, results, recommendations, profile, ingredient reference, about, and feedback workflows.
- **Backend:** FastAPI served by Uvicorn, SQLAlchemy ORM, and Alembic migrations. Authentication uses JSON Web Tokens, refresh-token versioning/revocation, and bcrypt password hashing.
- **Machine learning:** PyTorch for training and inference and MediaPipe [LTN+19] for face detection and landmark extraction.
- **Database:** SQLite for local development and PostgreSQL for staging through the same SQLAlchemy models.
- **Datasets:** ACNE04 for acne-severity training/evaluation and a filtered, modified cosmetics dataset for the product catalogue.
- **Tooling:** Git/GitHub, GitHub Actions, draw.io, and pytest.

2.11 Implementation Details

2.11.1 Module 1: Skin Analysis Pipeline

The skin analysis pipeline takes a raw uploaded photograph and produces a validated, classified skin analysis report used by the recommendation engine.

Photo Validation

Before classification, a MediaPipe-based validation gate checks face detectability, framing, and image quality. Invalid images are rejected with actionable feedback such as a missing face or unsuitable image quality.

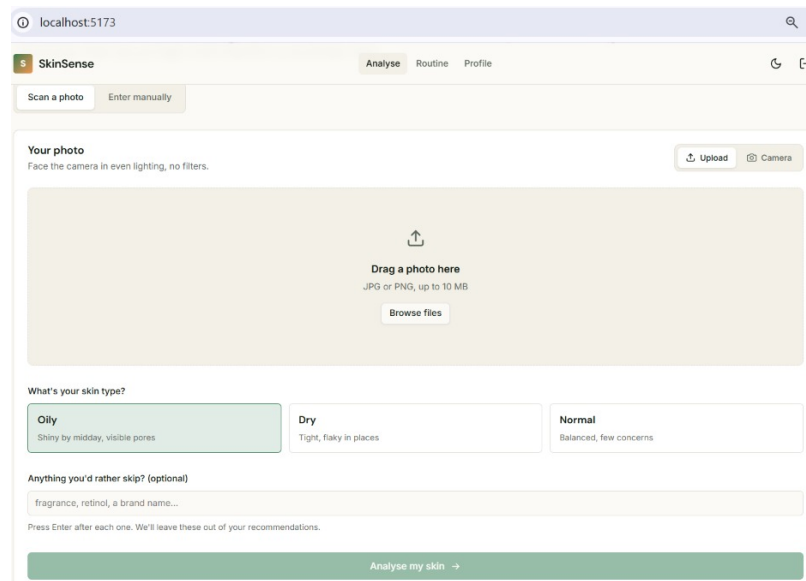


Figure 2.12: Photo upload interface showing the accepted file types and size limit, the entry point to the validation gate.

Acne Severity Classification

A ResNet-50 model classifies validated images into mild, moderate, and severe categories. The training pipeline uses a clean train/test split, class weighting, and augmentation. The current project target is at least 80% test accuracy.

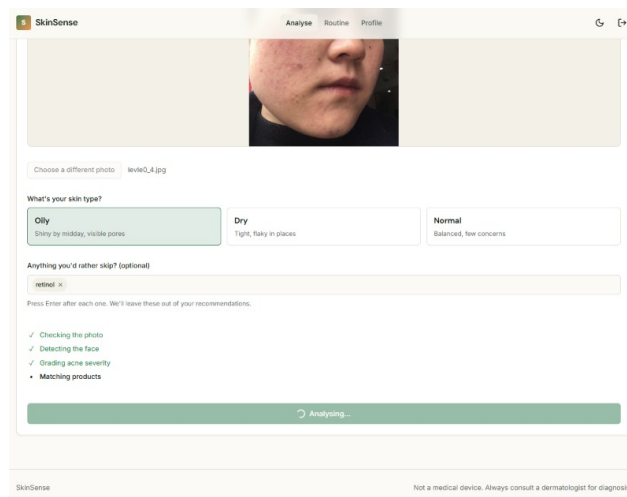


Figure 2.13: Live progress indicator showing the analysis stages – photo check, face detection, severity grading, and product matching.

Skin Type Classification

A MobileNetV2-based classifier is being developed to estimate skin type. The image model is complemented by an in-app questionnaire for cases where the image alone does not provide sufficient information.

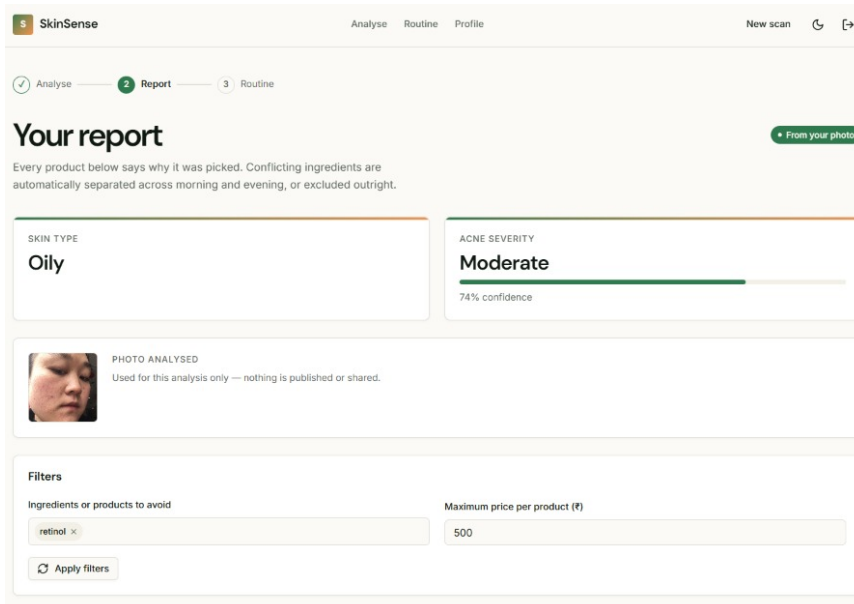


Figure 2.14: Generated report showing the classified skin type and acne severity together with the model's confidence score.

Referral Flag

The compiled report carries a dermatologist-referral flag when the result is severe or model confidence is below the configured threshold.

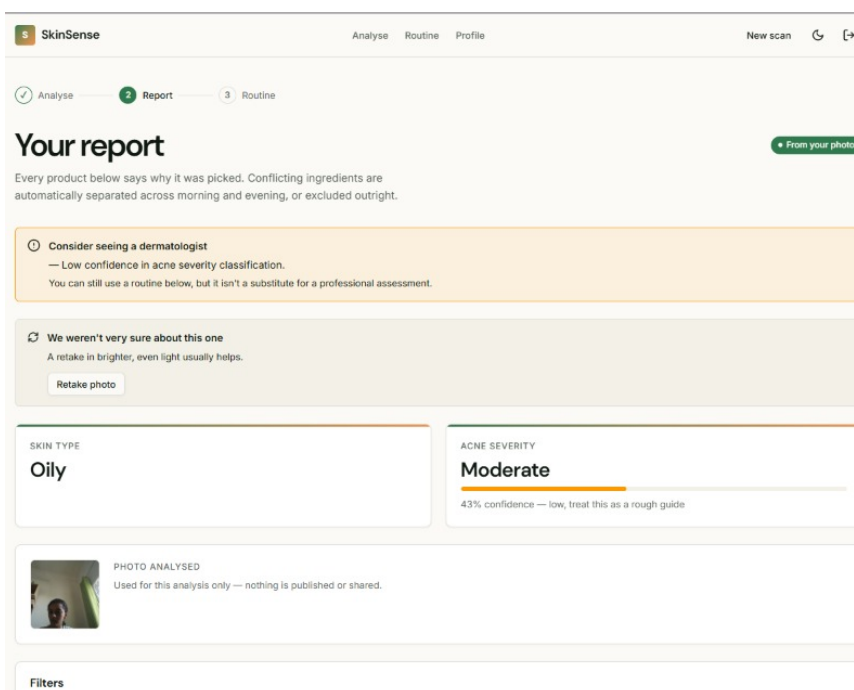


Figure 2.15: Report view with the dermatologist-referral notice, shown when classification confidence falls below the configured threshold.

Authentication and Access Control

JWT-based authentication protects endpoints used to submit photographs and retrieve analysis results so that user-specific information remains associated with the authenticated account.

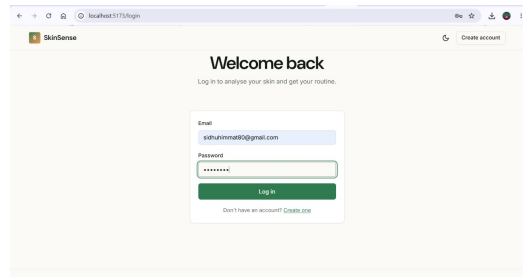


Figure 2.16: Login screen guarding access to analysis and profile endpoints via JWT-based authentication.

2.11.2 Recommendation Pipeline

The recommendation engine receives the skin analysis report together with the user's allergy list and budget limit. It retrieves candidate products from the catalogue, filters them by skin profile and severity, excludes allergenic and conflicting ingredients, applies the budget constraint, and assembles an AM/PM routine with explanations and safety information.

2.12 Diagram Coverage

For completeness, the design artefacts included in this report are:

1. Use Case Diagram
2. Use Case Tables (embedded PDF)
3. Activity Diagram
4. Swimlane Activity Diagram
5. Class Diagram
6. Collaboration Diagram
7. Sequence Diagram
8. Project Gantt Chart
9. DFD Level 0 – Context
10. DFD Level 1 – System Decomposition
11. DFD Level 2 – Skin Analysis Decomposition
12. DFD Level 2 – Recommendation Decomposition

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